# Data Context & Business Rules Register

Sep 28, 2026 · @Rajat

Everything we have learned about Ellementry's data and business that the data model depends on. Every rule here is either confirmed by the business or observed directly in the data, and says which. When a rule changes, update it here first, then in the model.

## 1. Company and business model

- **Ellementry** sells luxury home decor. Its parent company is **Dileep Essentials Pvt Ltd (DEPL)**, a manufacturer. The head office is in **Bagru**, which is also the main warehouse. *(confirmed)*
- The ERP location **"08 Jaipur Branch" is the Bagru warehouse**: the same physical place. *(confirmed)*
- Exports are routed through Dileep group entities and billed to them **cost-to-cost, with no profit**. *(confirmed)*
- Amazon FBA locations (DED4, Thane, Bengaluru 7, AMD2, BOM5 and others) are Amazon warehouses that hold Ellementry's own stock. *(confirmed)*

### Sales channels and how each is supplied

| Channel | How it works | Final or sale-or-return |
|---|---|---|
| Own website (Shopify) | D2C orders, prepaid (Easebuzz, Snapmint) or COD (Delhivery, Shiprocket) | Final, minus customer returns |
| Marketplaces | Amazon (FBA, Easyship, Seller Flex), Myntra, Nykaa Fashion, Tata Cliq, Pepperfry, FNP | Final, minus customer returns |
| Own stores | e.g. HYD Banjara Hills, Jaipur C-Scheme | Final, minus customer returns |
| B2B / wholesale | Outright invoices to business accounts | **Final** |
| Franchise stores | Given stock; return what doesn't sell | **Sale-or-return** |

| Channel | How it works | Final or sale-or-return |
|---|---|---|
| SIS (shop-in-shop) | Counters inside other retailers; return what doesn't sell | **Sale-or-return** |
| Quick commerce | Blinkit, Instamart; return what doesn't sell | **Sale-or-return** |
| Exports | Billed to Dileep group entities at cost | Final, ~zero margin |

In Domestic Sales, **only true B2B invoices are final revenue**. Franchise, SIS and quick-commerce invoices may later be reversed by returns. *(confirmed)*

## 2. Project scope and approach

- **End goal:** an AI hub where specialised agents (Executive, Marketing, Commerce, Product, Stock, Finance, Customer, Design, Operations) all read from one shared, clean data layer. The architecture was adapted from a consultancy proposal; the agency is not involved.
- **Stack:** Python + FastAPI backend, React frontend, PostgreSQL for structured data, pgvector (inside PostgreSQL) for semantic search instead of a separate vector database.
- **Prototype:** two agents first, **Stock AI and Operations AI**. The data layer is built before any agent.
- **Build:** one person. Local PostgreSQL on a laptop during development, moving to the cloud later, so SQL stays portable.
- **Standard:** this is company data going to production. Nothing is assumed: anything unclear is asked and recorded here.
- **Primary output:** executive dashboards and BI reporting, then agents on top.

## 3. Data collection rules

- **Period:** business dates **1 Apr 2025 to 15 Sep 2026** (about 1.5 years). The files shared so far are samples covering Apr to Sep 2026. *(confirmed)*
- **Cut-off:** some files run past 15 Sep (the return files go to 22 or 23 Sep). Anything after **15 Sep 2026** is excluded. *(confirmed)*
- **Method:** manual Excel exports from the ERP and from each platform's seller panel for now. Automation comes later, once the model is proven. *(confirmed)*

- **Inventory:** only the **current day's** stock snapshot can be pulled. Past days cannot be re-pulled, so history exists only from the day we start saving snapshots daily. *(confirmed)*
- **Product master:** the Ellementry Final Master is updated whenever something changes, so a fresh copy is taken with every data pull. *(confirmed)*

## 4. Confirmed business rules

| Area | Rule |
|---|---|
| Revenue | Headline sales value is **excluding GST** |
| Revenue | Net sales = invoices minus returns. Sale-or-return partners (franchise, SIS, quick commerce) can reverse invoices by returning unsold stock |
| Retail sales | Every retail invoice is a genuine D2C, marketplace or own-store sale |
| Retail sales | The sales channel comes from `CONTACTNAME` (e.g. Shopify_Customer = own website, including all COD and Snapmint orders) |
| Retail sales | FBA-named contacts (e.g. "Dileep Essentials Pvt Ltd FBA DED4") are Amazon customer orders fulfilled from that FBA warehouse |
| Retail sales | The bare contact "Dileep Essentials Pvt Ltd" (39 lines) is exports through Dileep |
| Retail returns | Every retail return is a genuine customer return. FBA-named contacts mean the return was received back at that Amazon FBA warehouse |
| Domestic sales | Only true B2B invoices are final. Franchise, SIS and quick-commerce billing names will be supplied so they can be separated |
| Domestic sales | Dileep group names (Dileep Crafts, Dileep Industries, Dileep Potteries, Dileep Jewellery) are exports, billed at cost. Low-value bills on these accounts are also at cost price |

| Area | Rule |
|---|---|
| Domestic sales | Any person's name (not a company) is an employee taking stock to exhibitions and bringing it back. No list needed: any name not on the business-account list is treated as an employee |
| Domestic returns | Contain B2B returns, unsold stock back from franchise, SIS and quick commerce (e.g. Kiranakart = Blinkit), employee exhibition returns, and stock from closed own stores |
| Purchase orders | The PO report also holds return invoices raised by companies and own stores sending stock back. A factory list will be supplied; any other counterparty is a return-inward, not a purchase |
| Products | Only **Trading** items are sellable SKUs. The inventory report also holds Packaging, Accessory and Stores & Consumables |
| Products | The Final Master (Sheet1) is the list of SKUs; **Continued** = currently sold, **Discontinued** = no longer sold |
| Locations | Bagru = main warehouse and head office. ERP "08 Jaipur Branch" = the Bagru warehouse |
| Dispatch (WMS) | B2C orders only, all dispatched from the Bagru warehouse |
| Quick commerce | Blinkit: **DELIVERED = sold**. Returns come in a separate sheet |
| Quick commerce | Manual export only for now |

## 5. Source file reference

All observations below come from the sample files (Apr to Sep 2026).

| Source file | What it is | One row per | Observed quirks |
|---|---|---|---|
| Retail Sales | D2C invoices: website, marketplaces, own stores, Amazon FBA | Invoice line | `CHANNELNAME` is the billing location, not the channel. Non-product lines: SHIPPING R, carry bags COCB04-06, Gifting. 144 Jaipur Branch rows have blank payment mode. Contains customer PII |
| Domestic Sales | B2B, franchise, SIS, quick commerce, group exports, employees | Invoice line | Mixed partner types in `CONTACTNAME`. 209 lines have blank `Taxable`. 2 Singapore rows |
| Retail Sales Return | Customer returns, incl. those received at Amazon FBA | Return line | `BILLNUM` links to the original sale (93% matched). SHIPPING R reversal lines. Runs past cut-off |
| Domestic Sale Return | B2B and sale-or-return partner returns, employee returns, closed-store stock | Credit-note line | Return date is `RTNFROMCUSTDT` (different name from retail); ~10% malformed, e.g. "08/04/2026;08/04/2026" |
| Stock Transfer (outward) | Stock sent between locations | Transfer line | Quantities negative. Sending location only in the document number prefix (JPR/, THA/, HYD, BAG/...). `CONTACTNAME` = receiving location in long form |
| Stock Transfer Inward | Stock received from a transfer | Receipt line | `REFDOCNUM1` links to the outward document (100% matched). 2 repeated document + SKU pairs. Some outward stock not yet received (in transit) |
| Pending Purchase Order | All PO lines for the FY, any status | PO line | Also holds return-inwards. Median lead time PO to goods receipt: 23 days (received POs in sample) |

| Source file | What it is | One row per | Observed quirks |
|---|---|---|---|
| Inventory report (.xls) | Stock on hand across 38 ERP locations, one day | SKU x location | Wide layout: repeating Qty + MRP blocks per location, 19 header rows. Includes non-trading items. Blinkit dark stores appear as ERP locations |
| Ellementry Final Master | Current SKU list with Continued / Discontinued | SKU | Only Sheet1 is structured. 472 Continued, 3,548 Discontinued |
| WMS dispatch | B2C order fulfilment from Bagru | Outbound delivery | Channel is in `Site Location`; `Customer` is a placeholder. Several SKUs in one cell. Long IDs in scientific notation. Contains PII |
| Blinkit sales | Consumer sell-out | Order line | Uses Ellementry SKU codes directly |
| Blinkit inventory | Stock at Blinkit feeder warehouses | Item x warehouse | Negative "net scheduled inventory" values |
| Instamart sales | Consumer sell-out | Date x store x item | Uses Ellementry SKU codes directly |
| Instamart inventory | Stock at Instamart facilities | Item x facility | |

## 6. Column traps

These produce wrong numbers without any error. Each is handled explicitly in staging, with a check that fails the load if the pattern ever changes.

| Trap | Correct handling |
|---|---|
| Retail `PRODASSESSABLEVALUE` **includes** GST (MRP minus discount), despite its name | Use `PRODDOCVALUE` as the excluding-GST value (verified: PRODDOCVALUE + GST = PRODASSESSABLEVALUE on every product line) |

| Trap | Correct handling |
|---|---|
| Domestic `Net Amount` **includes** GST | Use `Taxable` (verified: Taxable + Total Tax = Net Amount on 100% of lines) |
| `CHANNELNAME` in the sales files is a location, not a channel | Channel from `CONTACTNAME` |
| WMS `Customer` column is a placeholder (999999) | Channel from `Site Location` |
| WMS dates are month/day (4/14/2025); ERP text dates are day/month (08/04/2026) | Date format declared per source, never auto-detected |
| Long numbers exported from Excel lose digits (e.g. `1.60396E+13`) | Export IDs as text or CSV; load all IDs as text |
| Warehouse names differ by file ("24 FBA WH - AMD2" vs "Dileep Essentials Pvt Ltd Ahmedabad FBA AMD2") | One location code, with every spelling mapped to it |
| ERP location "27 Blinkit - Pune P2" and Blinkit's "Pune P2 - Feeder" report the same stock | Count each location once, from its system of record |
| Quick-commerce sell-out and ERP invoices describe the same goods | Never add them together: invoices = revenue, sell-out = consumer demand |
| Same document + SKU can repeat on separate lines | Line-occurrence counter in every key |

## 7. Still pending from the business

| Item | Needed for | Until then |
|---|---|---|
| Billing names of franchise stores, SIS and quick-commerce partners | Separating final B2B revenue from sale-or-return | Domestic accounts treated as B2B only once listed |
| Factory list | Separating purchases from return-inwards in the PO report | PO lines stay unclassified |

| Item | Needed for | Until then |
|---|---|---|
| Sell-out data from franchise and SIS stores (how to extract it) | True consumer demand for those channels | Demand for them uses net sales after returns |
| Blinkit return sheet (and Instamart's, if any) | Quick-commerce returns | Not loaded |
| Meaning of Blinkit's negative "net scheduled inventory" | Blinkit stock accuracy | Stored but excluded from stock numbers |
| Full 1.5-year data pull for every source | Loading the prototype | Samples only |

## Asks to the ERP team

- Add the branch / location column to the Stock Transfer, Stock Transfer Inward and Return exports.
- Export tracking, manifest and order numbers as text, or the whole report as CSV.
- Save the stock report every day from now on, plus Blinkit and Instamart inventory, into a dated folder.